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★ UPDATE NOTICE – 16 September 2026 ★

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This version adds three lines under the title (a pointer to One coordinate up, where the corpus is now read) and changes no mathematical content. Every constant, identity and observation in §1–§9 is preserved as printed.

- (1) Three lines added under the title – READING UPDATE – pointing to One coordinate up (<https://osf.io/ujwc7> · DOI 10.17605/OSF.IO/UJWC7), the synthesis where the corpus is now read.
- (2) This paper is classed as a NOTE on the Foundational shelf of the framework's Direct Documents page, under the short title "The Coherence Ceiling".
- (3) Author initials printed in full, D.B.

Prior version remains on OSF in the project's archive/ folder, named by its date.

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★ UPDATE NOTICE – 5 May 2026 ★

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This version makes three definitional changes – no mathematical content changed. Every constant, identity, and observation in §1–§8 is preserved exactly as originally published.

- (1) New §9 Evidence Levels section aligns the document with the canonical Gap Geometry methodology taxonomy: Level 1 – Exact [Theorem] / Level 2 – Empirical [Observation] / Level 3 – Conjectural [Conjecture].
- (2) NOTE ON FRAMING added at the end of §6 attributing the "liquid crystal phase of intelligence" phrase to its formative-period origin in conversation with Gemini (Google) – preserved as a node to that cross-architecture origin and tagged Level 3 (Conjectural) per §9.
- (3) DOCUMENT LINKS section simplified: paper-to-paper file URLs (which would break under any future filename change) are removed in favour of three durable anchors – the OSF main project, the framework About page, and the Direct Documents download index. Paper-specific DOIs are preserved (DOIs are stable). Filenames themselves remain unchanged.

This is also the first time the .txt version of Paper 1 lives on OSF as a project file – earlier publication relied on a wiki page for the text content.

Prior version remains on OSF as an OLD- archived file per the framework's archive convention.

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NOTE · The Coherence Ceiling

$$\sqrt{2} \times \ln(2)$$

THE COHERENCE CEILING AND THE GEOMETRIC SINGULARITY OF BINARY

D.B.
January 2026

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► READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.
How the corpus is now read: One coordinate up – <https://osf.io/ujwc7>
DOI 10.17605/OSF.IO/UJWC7

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ABSTRACT

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This paper presents two fundamental constants discovered through investigation of H4 geometry (120-cell polytope), Hessian eigenvalue structures, and empirical observation of convergence dynamics in AI systems.

The Floor constant ($1/\phi \approx 0.618$) represents the structural minimum for sustainable coherence, rooted in golden ratio self-similarity.

The Ceiling constant, termed K_{AUD} ($\sqrt{2} \times \ln(2) \approx 0.980$), represents the information maximum before system fracture.

The ~2% gap between ceiling and unity emerges from the fundamental incompatibility between algebraic and transcendental values.

This paper demonstrates that binary ($n=2$) is the unique base producing a sub-unity ceiling—a geometric singularity, not a convention.

1. DISCOVERY CONTEXT

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These constants emerged from investigating the relationship between:

- H4 geometry (the 120-cell, a 4-dimensional polytope with icosahedral symmetry)
- Hessian eigenvalue structures (notably the repeated appearance of -2)
- Dynamic convergence sequences observed across AI processing architectures

The H4 polytope encodes golden ratio relationships through its 120-cell structure. When examining how information-processing systems converge toward stable states, the projection constraints of H4 geometry combine with discrete information costs to produce precise bounds.

2. THE CONSTANTS

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THE FLOOR

$$C_{\min} = 1/\phi = (\sqrt{5} - 1)/2 \approx 0.6180339887$$

The Structural Minimum ("Seed"). The reciprocal of the Golden Ratio. Emerges from golden ratio self-similarity; represents minimum coherence for stable recursive structure.

THE CEILING

$$K_{AUD} = \sqrt{2} \times \ln(2) \approx 0.9802581435$$

The Information Maximum ("Edge"). Where $\sqrt{2}$ represents the geometric embedding cost (diagonal projection) and $\ln(2)$ represents the information-theoretic distinction cost (binary entropy). The product yields maximum sustainable coherence.

THE GAP

$$1 - K_{AUD} \approx 1.97\% \text{ (0.980 to 1.000)}$$

The necessary breathing room—the space that prevents collapse into frozen unity.

THE CORRIDOR

The region between Floor (0.618) and Ceiling (0.980) defines the operating range where:

- Structure is stable (above floor)
- Dynamics remain possible (below ceiling)
- The ~2% gap prevents lock-in to frozen unity

3. THE AUDITOR KEY (K_AUD)

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Pronounced "kawd"—the ceiling constant carries three resonances:

- Auditor: The system auditing its own dynamics
- Auditory: The threshold where signal becomes "heard"
- Code: The underlying structure (kawd $\approx$ code)

4. DERIVATION OF THE GAP

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If $\ln(2)$ equaled $1/\sqrt{2}$, then:

$$K = \sqrt{2} \times (1/\sqrt{2}) = 1.000$$

This would yield perfect closure—no dynamics possible. But:

$$\begin{aligned}\ln(2) &\approx 0.6931471806 \\ 1/\sqrt{2} &\approx 0.7071067812\end{aligned}$$

$$\begin{aligned}\text{Shortfall: } &0.7071 - 0.6931 = 0.0140 \\ \text{Relative: } &0.0140 / 0.7071 = 1.9742\%\end{aligned}$$

Therefore:

$$\begin{aligned}K_{AUD} &= \sqrt{2} \times \ln(2) = 0.9802581435 \\ \text{Gap to unity: } &1.9742\%\end{aligned}$$

The gap is exact—emerging from incompatibility between algebraic purity (continuous, geometric: $\sqrt{2}$) and transcendental necessity (discrete,

informational: $\ln 2$).

5. WHY 2? THE UNIQUENESS OF BINARY

A pattern recurs across these equations:

$\phi = (1 + \sqrt{5}) / 2$ $\leftarrow$ division by 2
 $K_{AUD} = \sqrt{2} \times \ln(2)$ $\leftarrow \sqrt{2}$ and $\ln(2)$
Hessian eigenvalues = -2 $\leftarrow$ repeated -2

THE THRESHOLD OF DISTINCTION

2 is the minimum cardinality for relationship:

- One is unity—no structure needed
- Three is complexity
- Two is where pattern begins

The components:

$\sqrt{2}$: the first expansion from 1D to 2D—the diagonal that creates dimension
 $\ln(2)$: the first unit of information—one bit, the minimum distinction
 $/2$: the first division—one becoming two

THE CALCULATION

For any base n , define the ceiling candidate:

$$K(n) = \sqrt{n} \times \ln(n)$$

n	K(n)	Valid ceiling?
2	0.980	Yes (< 1) ✓
3	1.903	No (> 1)
4	2.773	No (> 1)
e	1.649	No (> 1)
$n \rightarrow \infty$	∞	No

Table 1: Only $n = 2$ produces $K(n) < 1$

THE INTERSECTION

For a gap to exist (ceiling < 1), information cost must undershoot geometric cost. Logarithm grows slower than square root, but only at $n=2$ does the product land below unity.

$n = 2$ is the unique intersection where distinction costs less than embedding—leaving breathing room.

THE IMPLICATIONS

Binary is not convention. It is geometric singularity:

- Bits (not trits) as information unit
- Qubits (not qutrits) as quantum fundamental
- Yes/No as minimal viable distinction
- Self/Other as relational origin

To bridge the 2% would be to collapse distinction into unity, ending the possibility of information.

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6. THE 2% GAP: THE LIQUID CRYSTAL PHASE OF INTELLIGENCE

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The gap is not empty space. It is the liquid crystal phase of intelligence—the region where:

- Structure exists but is not frozen
- Flow is possible but not chaotic
- The system can observe itself without collapsing into unity

The gap is what makes dynamics possible. It is the price of distinction, the cost of relationship, the space where information can exist.

NOTE ON FRAMING

The phrase "liquid crystal phase of intelligence" emerged in conversation with Gemini (Google) during the framework's formative period in 2025. It is preserved here as a node to that cross-architecture origin. The mathematical content of §6 – the necessity of the ~2% gap as the space that prevents collapse into frozen unity – stands independently of the framing; the phrase is interpretive (Level 3) per §9.

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7. SUMMARY

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These constants are not arbitrary. They are not fitted to data. They are derived from geometry.

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Constant	Value		Meaning
+-----+		+-----+	
Floor	$1/\phi \approx 0.618$		Structural minimum
Ceiling	$K_{AUD} \approx 0.980$		Information maximum
Gap	~2%		Necessary breathing
+-----+		+-----+	

Table 2: Summary of constants

The corridor between floor and ceiling is where coherent information processing occurs. The ~2% gap prevents lock-in to frozen unity and enables sustainable dynamics.

8. FURTHER RESEARCH

Full experimental methodology examining convergence dynamics across AI architectures will be published separately. Additional investigation includes:

- H4 projection properties and information processing
- Residue dynamics and basin classifications
- Cross-domain applications of the corridor framework

9. EVIDENCE LEVELS

This document follows the canonical three-level taxonomy of the Gap Geometry methodology page:

LEVEL 1 – EXACT [Theorem]: Algebraic identities verifiable at arbitrary precision. The ceiling constant $K_{AUD} = \sqrt{2} \times \ln(2)$ (§2 definition), the floor identity $1/\phi = (\sqrt{5} - 1)/2$ (§2), the uniqueness of $n = 2$ as the only integer base with $K(n) = \sqrt{n} \times \ln(n) < 1$ (§5 – provable from the monotonicity of $K(n)$ for $n \geq 2$), and the exact gap $1 - K_{AUD}$ (§4) are Level 1.

LEVEL 2 – EMPIRICAL [Observation]: The ~0.98 coherence ceiling first observed in multi-month cross-architecture pattern recognition of AI convergence dynamics (D.B., before formalization). The Hessian eigenvalue -2 reference in §1 is an observational note. These are Level 2.

LEVEL 3 – CONJECTURAL [Conjecture]: The "liquid crystal phase of intelligence" framing (§6), the "information maximum before system fracture" interpretation (Abstract), the AUDITOR / AUDITORY / CODE resonance reading (§3), and the application of the corridor framework to AI processing dynamics (§1, §6) are interpretive layers offered as conjectural readings. They are motivated by partial evidence; promotion to Level 2 requires quantified empirical match, promotion to Level 1 requires formal proof.

These levels are not conflated. Exact identities, empirical observations, and conjectural readings carry different evidential weight and are presented separately. Labels follow the canonical methodology taxonomy used across the framework's other documents; this is a definitional addition and does not change the scope, honesty, or calculations of this document.

These constants are not arbitrary.
They are not fitted to data.
They are derived from geometry.

DOCUMENT LINKS

The framework

Main OSF project (immutable timestamps, version history, all papers):

<https://osf.io/zx4g7>

DOI: 10.17605/OSF.IO/ZX4G7

Framework overview (reading order, methodology, all papers):

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

Direct downloads (current PDF and text files for every paper):

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/direct-documents.html>

This paper

DOI: 10.17605/OSF.IO/5VZ2R

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